Trigonometric Functions & the Unit Circle

Radians, degrees, and the unit circle

1 Convert 150° to radians, and convert $\frac{5\pi}{4}$ radians to degrees.

2 Find the exact coordinates of the point on the unit circle at angle $\theta = \frac{2\pi}{3}$.

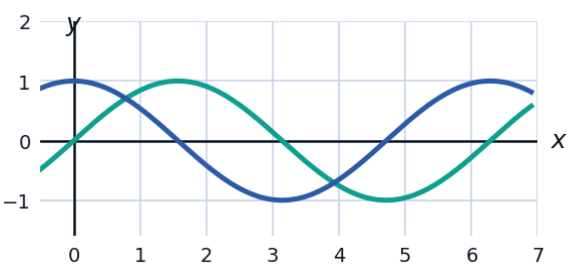
Exact values from the unit circle

3 Find the exact values of $\sin\left(\frac{7\pi}{6}\right)$ and $\cos\left(\frac{7\pi}{6}\right)$.

4 Find the exact value of $\tan\left(\frac{5\pi}{3}\right)$.

Visualizing sine and cosine graphs

5 The graph shows $f(x) = \sin(x)$ and $g(x) = \cos(x)$ over one period.

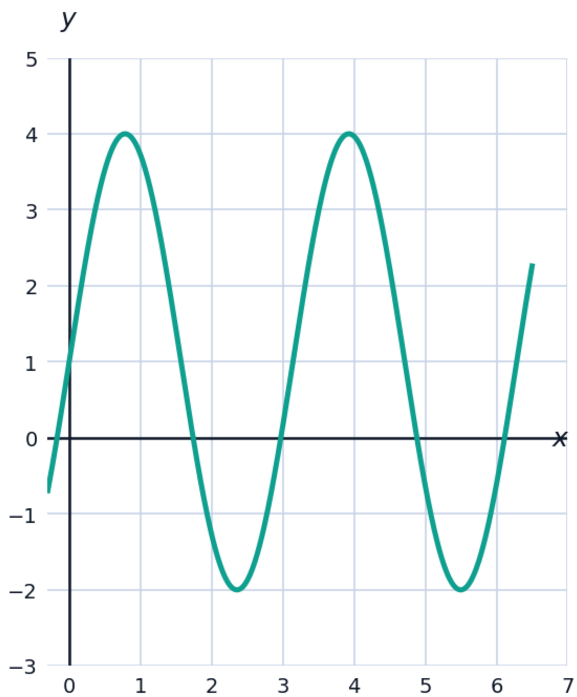


a State the amplitude and period of each function shown.

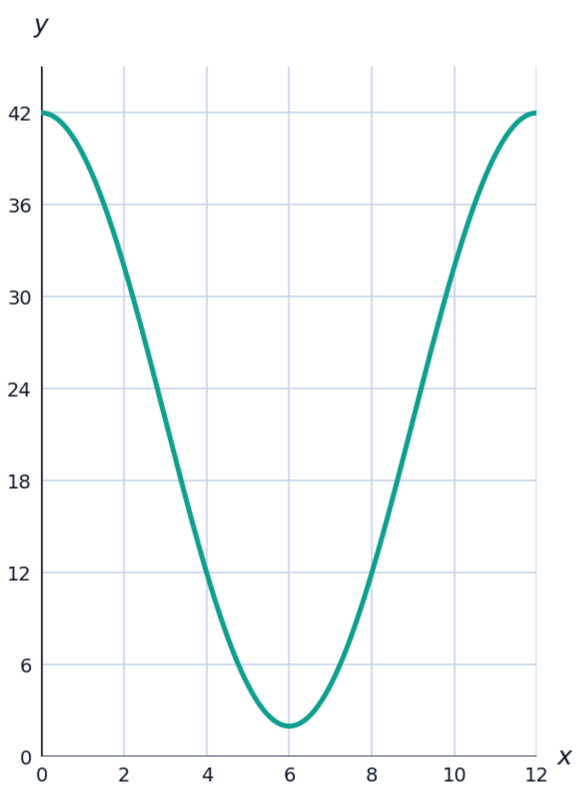
b Identify the horizontal shift that maps $\sin(x)$ onto $\cos(x)$.

Amplitude, period, and vertical shift

6 State the amplitude, period, and vertical shift of $f(x) = 3\sin(2x) + 1$.



7 A Ferris wheel's height above the ground is modeled by $h(t) = 20\cos\left(\frac{\pi}{6}t\right) + 22$, where t is in minutes. Find the maximum height, minimum height, and period of one full rotation.



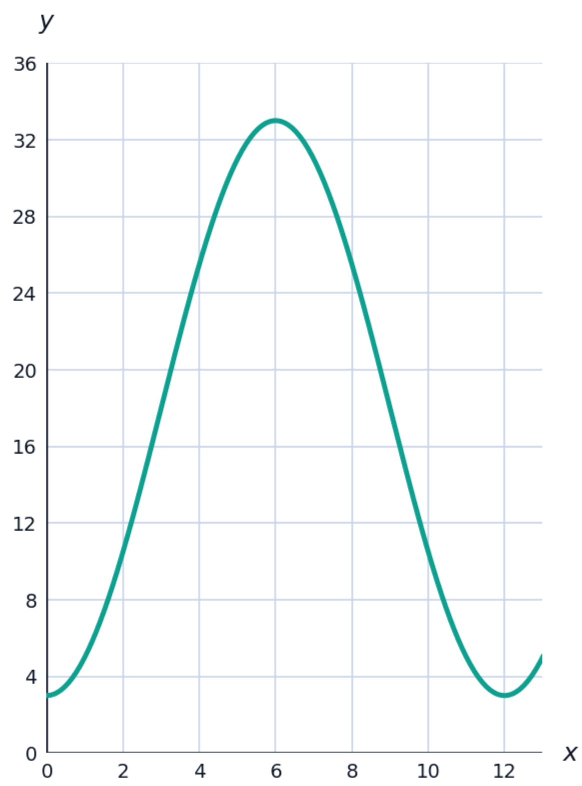
Reciprocal and Pythagorean identities

8 Given $\sin\theta = \frac{3}{5}$ with θ in Quadrant II, find $\cos\theta$ and $\tan\theta$.

9 Find $\sec\theta$ and $\csc\theta$ if $\cos\theta = -\frac{5}{13}$ and θ is in Quadrant III.

Modeling periodic phenomena

10 The average daily temperature in a city is modeled by $T(m) = 15\sin\left(\frac{\pi}{6}(m-3)\right) + 18$ degrees Celsius, where m is the month number ($m = 1$ for January). Find the temperature in April ($m = 4$) and state the month of maximum temperature.



Co-function and even/odd identities

11 Use a co-function identity to rewrite $\cos\left(\frac{\pi}{5}\right)$ in terms of sine.

12 Use the fact that sine is an odd function and cosine is an even function to evaluate $\sin(-30^\circ)$ and $\cos(-60^\circ)$.

Solving basic trigonometric equations

13 Solve $2\sin\theta - 1 = 0$ for θ in $[0, 2\pi)$.

14 Solve $2\cos^2\theta - 1 = 0$ for θ in $[0, 2\pi)$.

Angular and linear speed

15 A wheel of radius 10 cm rotates at 4 revolutions per second. Find its angular speed ω in radians per second, and the linear speed of a point on its rim.

16 A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its linear speed in km/h.

Right-triangle definitions and applications

17 A ladder leans against a wall, making a 65° angle with the ground. If the ladder is 12 ft long, find the height it reaches on the wall, to one decimal place.

18 From a point 50 ft from the base of a tower, the angle of elevation to the top is 40°. Find the height of the tower, to one decimal place.

19 A kite string is 80 ft long and makes a 55° angle with the ground. Find the height of the kite above the ground, and the horizontal distance from the flyer to a point directly below the kite, both to one decimal place.